

Tumors of central nervous system

Classification of intracranial and spinal cord tumors:

Cell of Origin	Tumour designation
Glial cells	Gliomas: -Astrocytoma - Oligodendroglioma -Ependymoma
Choroid plexus	-Choroid plexus papilloma
Neurons	-Neuroblastoma - Ganglioneuroblastoma -Ganglioneuroma
Embryonic/Primitive cells	- Medulloblastoma -Glioblastoma
Meningothelial cells	Meningioma
Nerve sheath	-Neurofibroma -Schwannoma -Malignant nerve

	sheath tumor
Blood vessels	- Haemangioblastoma
Miscellaneous	-Lymphoma - Craniopharyngioma -Pituitary tumors -Pineoblastoma - Metastatic tumors

Epidemiology:

- Incidence: 8-25/ 100.000
- Primary CNS tumors: They represent $\frac{1}{2}$: $\frac{3}{4}$ of all body tumors
- Metastasis : They represent $\frac{1}{4}$ of all tumors
- 20% of all CNS tumors Occur in childhood and are infratentorial in site (brain stem, cerebellum)
- Etiology: genetic, hereditary, viral infection (EB), Chemicals.

Brain metastases:

- They occur when tumour cells spread from their original site to the brain.
- About $\frac{1}{4}$ of brain tumors are metastatic. They may be:

- 1. Carcinomas: Lung (NSCLC is common), breast, kidney, and gastrointestinal tract carcinomas are the commonest.
- 2. Sarcomas.
- 3. Other malignancies include melanoma, lymphoma, and leukemia.
- The presence of a solitary brain metastasis can be confused with a primary brain tumor, and distinction is critical for selecting appropriate therapy.
- Brain biopsy is often required to establish a specific diagnosis.

World Health Organization (WHO) Brain Tumor Grades			
	Grade	Characteristics	Tumor Types
Low Grade	WHO Grade I	• Least malignant (benign) • Possibly curable via surgery alone • Non-infiltrative • Long-term survival • Slow growing	Pilocytic astrocytoma Craniopharyngioma Gangliocytoma Ganglioglioma
	WHO Grade II	• Relatively slow growing • Somewhat infiltrative • May recur as higher grade	"Diffuse" Astrocytoma Pineocytoma Pure oligodendroglioma
High Grade	WHO Grade III	• Malignant • Infiltrative • Tend to recur as higher grade	Anaplastic astrocytoma Anaplastic ependymoma Anaplastic oligodendroglioma
	WHO Grade IV	• Most malignant • Rapid growth, aggressive • Widely infiltrative • Rapid recurrence • Necrosis prone	Glioblastoma Multiforme (GBM) Pineoblastoma Medulloblastoma Ependymoblastoma

Gliomas:

- Gliomas are tumours of neuroglial cells that are, astrocytes, oligodendrocytes, and ependymal cells.

1. Astrocytomas

- These are gliomas composed predominantly of astrocytes.
- They are the most common primary tumours of CNS.
- There are 4 histological grades according to WHO.

WHO Tumour designation	
Grade I	Pilocytic astrocytoma
Grade II	Fibrillary astrocytoma
Grade III	Anaplastic astrocytoma
Grade IV	Glioblastoma

Pilocytic Astrocytoma (WHO Grade I)

- Age: They are usually seen in childhood and adolescence, hence known as juvenile pilocytic astrocytoma (JPA).
- Site: - typically involve midline structures:
 - (a) The optic chiasma & hypothalamus.
 - (b) The cerebellum.

Pathological changes:

- Gross picture: -They are well circumscribed and often cystic mass with a solid mural nodule.
- Microscopic picture: They are consisting of elongated hair-like (pilo) cells with elongated bipolar processes and contain Rosenthal fibers, which are dense, eosinophilic fibers within cytoplasmic processes of astrocytes. The nuclei are bland, oval to elongated.

Fibrillary Astrocytomas:

- Fibrillary astrocytomas are static or progress slowly.
- They include Well differentiated “diffuse astrocytoma” (WHO Grade II)
- Age: Usually in 4th to 6th decade of life (adulthood).
- Site: Commonly in cerebral hemisphere.

Pathological changes:

- Gross picture: -They appear as an ill –defined infiltrative growth.
- Microscopic features: It is consisting of elongate or irregular, hyperchromatic nuclei. The cytoplasm varies from indistinct to apparent as processes. The cellularity is Moderate with variable nuclear pleomorphism.

Anaplastic astrocytoma (WHO Grade III):

- Age & Site:
- -It occurs mainly in the cerebral hemispheres of adults.
- Gross picture: They appear as an infiltrative growth.
- Microscopic picture: has the following features:
- 1. More cellular.
- 2. Greater nuclear pleomorphism.
- 3. Presence of mitotic activity.

Glioblastoma multiforme (WHO Grade IV):

- This is a highly malignant tumor, usually causing the death of the patient within a year.
- Age & Site: It occurs mainly in the hemispheres of adults.

- Gross picture: Producing a well circumscribed mass with much necrosis and haemorrhage.
- Microscopic picture: are extremely pleomorphic, contain characteristic foci of necrosis and show glomeruloid vascular proliferation.

2. Ependymoma:

- Arise from ependymal cells lining ventricles or spinal cord. Ependymomas are usually less aggressive than astrocytomas
- Age: Ependymoma occurs mainly in childhood and adolescence.
- Site:
- -In childhood, it may occur in any part of the ventricular system, but especially the 4th ventricle.
- -In adults, the spinal cord is their most common location. The ependymoma is also the commonest glioma to involve the spinal cord.

Pathological features:

- Gross picture: circumscribed mass arising in relation to ventricles.
- Microscopic picture:
 - A specific histological variant (myxopapillary ependymoma, WHO grade I), occurs in the region of cauda equine, showing a prominent papillary growth pattern with myxoid degeneration of the stroma.(WHO grade II) show growth of ependymal cells around blood vessels (pseudorosettes) or around small spaces (true rosettes).
 - Anaplastic ependymoma (WHO grade III) shows increased cell

density, high mitotic rates, and necrosis.

3. Oligodendroglioma:

- It is derived from oligodendroglial cells (which provide myelin sheaths for axons in the CNS), usually arise in the white matter.
- - It is uncommon, slowly growing glioma.
- - These tumours are prone to spontaneous haemorrhage.
- - A characteristic feature is the presence of calcification, which can be seen radiologically.
- Age: - occurring in adults (4-5th decades).
- Site: - cerebral hemispheres, in white matter.
- Gross picture: It shows a circumscribed solid or gelatinous tumor.
- Microscopic picture: It shows numerous small uniform cells having rounded lymphocyte like nuclei and pale staining to clear cytoplasm (cytoplasmic halo “fried egg” appearance)
- Oligodendrogliomas have a substantially better prognosis than astrocytomas

Meningiomas:

- Are tumors of meningeal coverings of the brain, are mostly benign.
- Age: 30-60 years. More in females
- Site: hemispheres convexity, spinal canal.
- Gross picture: firm, lobulated mass, circumscribed and are always clearly separate from brain parenchyma without invading it, making it possible to completely resect these tumors with minimal loss of neurologic function.

- Microscopic picture: Most meningiomas are low grade, and mitotic activity is somewhat predictive of clinical behavior.
- They can have a variety of different microscopic patterns including fibrillary, psammomatous (numerous psammoma bodies are present with spindled tumor cells), and meningotheliomatous. None of these patterns is prognostically significant.
- Prognosis & behavior: Meningiomas grow slowly, displacing brain tissue, and many present with seizures.
- Some meningiomas may induce bone formation in overlying skull, resulting in a characteristic radiographic presentation.
- rare turn into malignant (meningeal sarcoma)

Retinoblastoma and Medulloblastoma:

- Both are rare, small, blue round-cell tumors of childhood.
- Retinoblastoma is a rare tumor of retinal epithelial cells that almost always occurs in early childhood and usually presents with unilateral visual loss and leukocoria (white reflex to light shined on the retina).
- Retinoblastomas confined to the orbit are cured by radical surgery (orbital exenteration).
- Familial cases with bilateral tumors may be treated with irradiation. Patients with extension beyond the orbit or with distant metastases usually have a poor response to therapy.
- Retinoblastoma has played an important role in understanding the role of tumor suppressor genes in tumor development.
- Medulloblastomas are malignant tumors of the midline cerebellum in children that are aggressive neoplasms and may metastasize outside the CNS.

Tumors of the peripheral nervous system:

They usually present as mass lesions rather than with symptoms related to neuronal dysfunction.

- When occurs in parenchymal organs, they may be confused with those primary tumors, or with sarcomas when they arise in soft tissue.
- Anatomic attachment to a nerve is a diagnostic clue.
- Most neural tumors express S100 protein, which is a useful immunohistochemical marker although it is expressed in other tumors (e.g., melanoma).

1- Schwannomas:

- Are tumors of the Schwann cells that produce the myelin sheaths about peripheral nerves.
- They are benign, curable and common in adults.
- Gross Picture: Firm, grayish, capsulated masses at one side of the nerve. They are sharply circumscribed from surrounding tissue.
- Microscopic picture: They usually have a characteristic morphologic appearance with fibrillary tumor cells palisading in parallel arrays (so-called Antoni A) and haphazardly arranged (Antoni B areas).
- Can cause neural dysfunction, and schwannomas of the eighth nerve (acoustic neuromas) typically present with hearing loss.
- Acoustic neuromas may be sporadic or associated with type II neurofibromatosis.
- Deep-seated schwannomas in patients with genetic neurofibromatosis are more likely to undergo malignant degeneration than are sporadic schwannomas.

Neurofibromas:

- 1- Sporadic: most often, in the skin and mucous membranes as poorly

circumscribed lesions.

- 2- Genetic: multiple neurofibromas often have genetic neurofibromatosis.
- Site: orbit, neck, back, inguinal region.
- Gross picture: They are not capsulated, along the affected nerve.
- Most sporadic neurofibromas are small and do not produce significant clinical symptoms.
- Genetic neurofibromatosis: develop many neurofibromas, large size, cause significant dysfunction and deformity.
- Microscopic picture:
 - They are composed of spindle cells with elongated wavy nuclei. Mast cells are usually prominent in neurofibromas and can be helpful diagnostically.
 - Patients with neurofibromatosis type 1 have mutations in the NF1 tumor suppressor gene.
 - These patients tend to develop numerous plexiform neurofibromas and schwannomas that have an increased risk for malignant degeneration (particularly in deep-seated tumors).
 - Affected patients often have café-au lait spots (flat pigmented lesions of the skin), which may suggest the diagnosis.
- Patients with neurofibromatosis type II have mutations of the merlin gene and have a much different constellation of findings, with acoustic neuromas, meningiomas, and characteristic spinal neurofibromas.